



Proton NMR Spectroscopy

Video Series 2

Video 4: Long Range Coupling in Aromatic Systems





Video Series Goals

Video 1: NMR Spectra of Mixtures

- Diagnostic NMR signals
- Calculating Percent Composition of a Mixture

Video 2: NMR Solvents (**Current**)

- Deuterated Solvents
- Solvent Signals in NMR

Video 3: Splitting Pattern Complexity

- Extension of “n+1” Rule

Video 4: Long Range Coupling (**Current**)

- 4-Bond (4J) Coupling

Video 5: Diastereotopic Protons

- 2-Bond (2J) Coupling

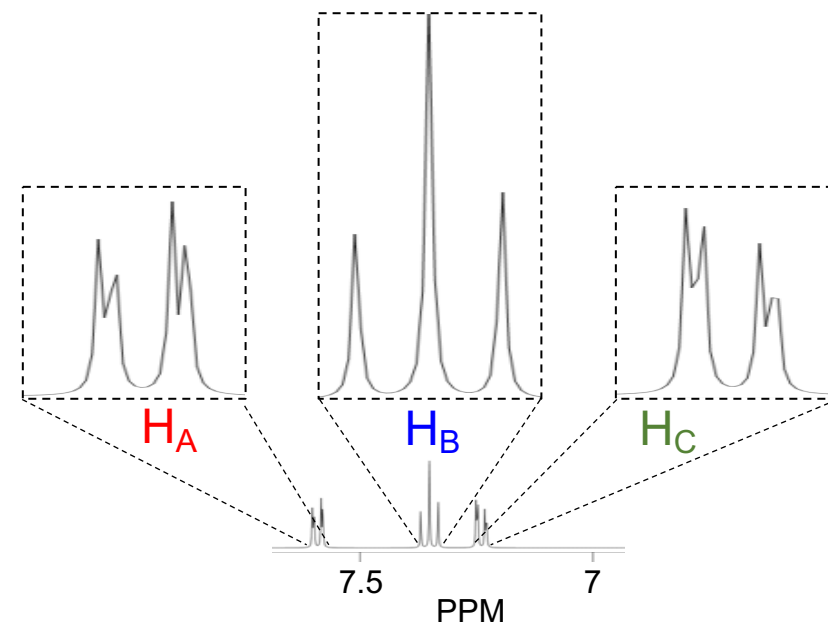
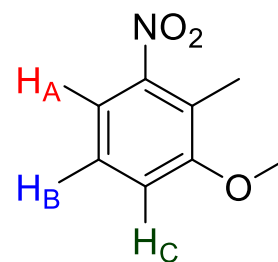
Long Range Coupling (4J)

Long range coupling occurs across **4** or more bonds

- Common in **aromatic** and other π systems (alkynes and allenes)
- Can occur in sp^3 systems with specific geometries (less common)

4J couplings are commonly visible in aromatic systems.

- $^4J_{\text{aromatic}} \sim 1\text{-}3 \text{ Hz}$ (usually 1.5 Hz)
- $^3J_{\text{aromatic}} \sim 6\text{-}9 \text{ Hz}$ (usually 7.5 Hz)



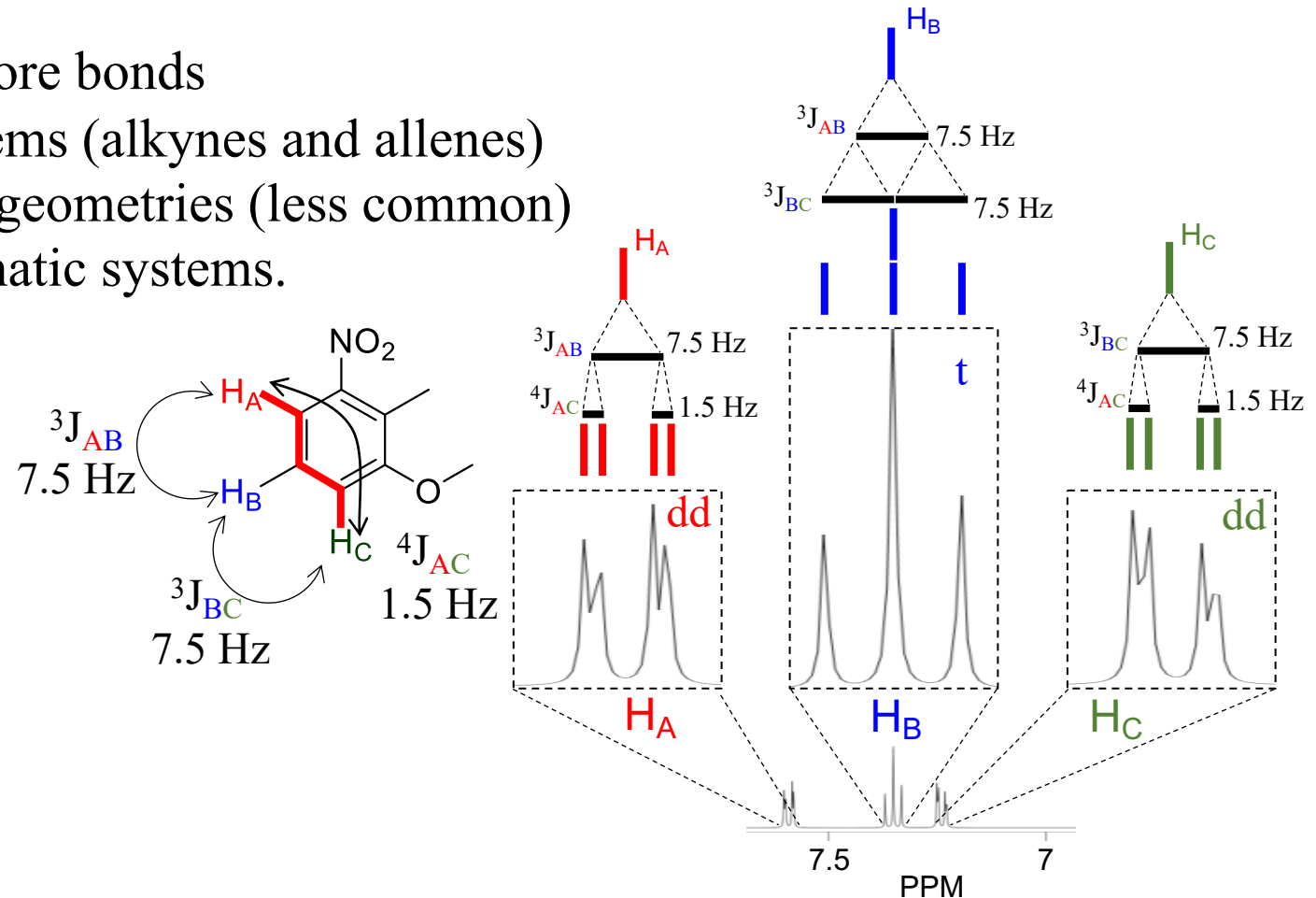
Long Range Coupling (4J)

Long range coupling occurs across **4** or more bonds

- Common in **aromatic** and other π systems (alkynes and allenes)
- Can occur in sp^3 systems with specific geometries (less common)

4J couplings are commonly visible in aromatic systems.

- $^4J_{\text{aromatic}} \sim 1\text{-}3 \text{ Hz}$ (usually 1.5 Hz)
- $^3J_{\text{aromatic}} \sim 6\text{-}9 \text{ Hz}$ (usually 7.5 Hz)





Aromatic Splitting patterns

Generally observe that all $^3J_{\text{aromatic}}$ coupling constants are equal

Generally observe that all $^4J_{\text{aromatic}}$ coupling constants are equal

- Exceptions exist, but this often holds true

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ **is always true**

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”

Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
---------------------------	---------------------------	---------------------------	-------------------------------

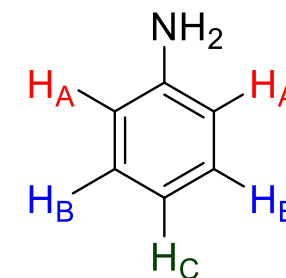
Aromatic Splitting patterns

Generally observe that all $^3J_{\text{aromatic}}$ coupling constants are equal
 Generally observe that all $^4J_{\text{aromatic}}$ coupling constants are equal
 • Exceptions exist, but this often holds true

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”



	Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
H_A	Shielded	1	1	dd
H_B	Neutral			
H_C	Shielded			

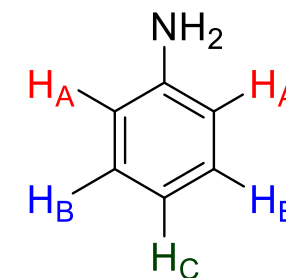
Aromatic Splitting patterns

Generally observe that all $^3J_{\text{aromatic}}$ coupling constants are equal
 Generally observe that all $^4J_{\text{aromatic}}$ coupling constants are equal
 • Exceptions exist, but this often holds true

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”



	Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
H_A	Shielded	1	1	dd
H_B	Neutral			
H_C	Shielded			

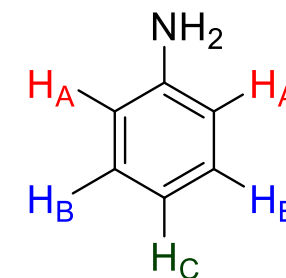
Aromatic Splitting patterns

Generally observe that all $^3J_{\text{aromatic}}$ coupling constants are equal
 Generally observe that all $^4J_{\text{aromatic}}$ coupling constants are equal
 • Exceptions exist, but this often holds true

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”



	Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
H_A	Shielded	1	1	dd
H_B	Neutral	2	0	t
H_C	Shielded	2	2	tt

Aromatic Splitting patterns

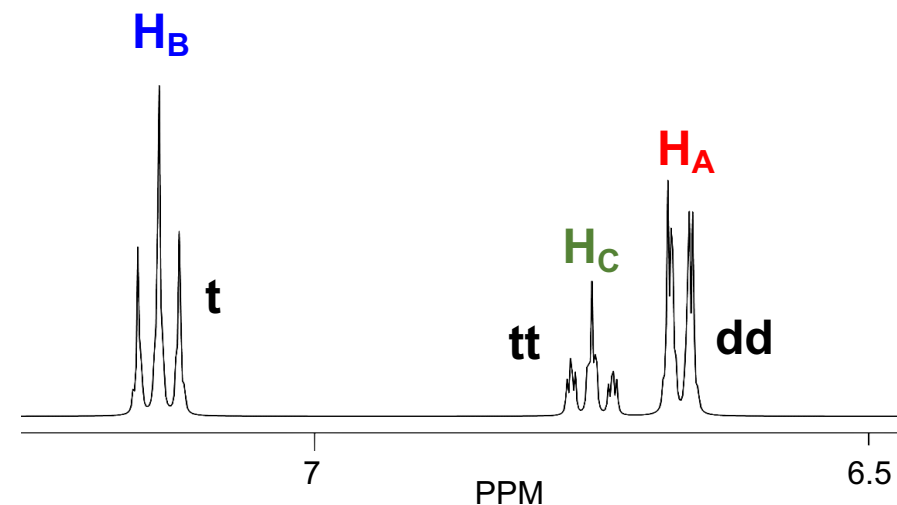
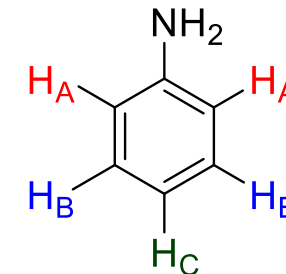
Generally observe that all $^3J_{\text{aromatic}}$ coupling constants are equal
 Generally observe that all $^4J_{\text{aromatic}}$ coupling constants are equal
 • Exceptions exist, but this often holds true

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”

	Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
H_A	Shielded	1	1	dd
H_B	Neutral	2	0	t
H_C	Shielded	2	2	tt

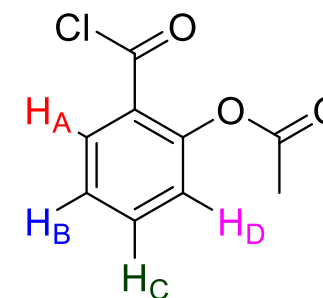


Aromatic Splitting patterns

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”



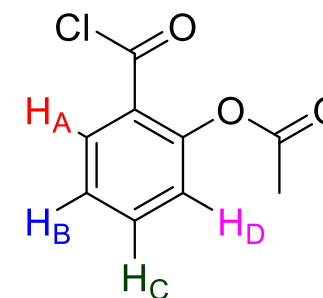
	Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
H_A	Deshielded	1	1	dd
H_B	Shielded	2	1	td
H_C	Deshielded	2	1	td
H_D	Shielded	1	1	dd

Aromatic Splitting patterns

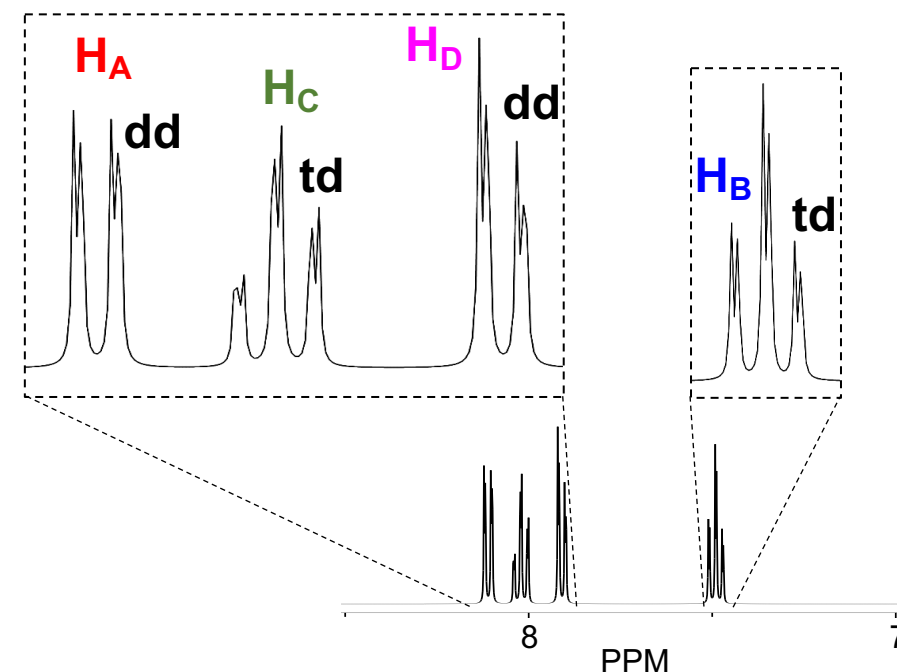
$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”



	Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
H _A	Deshielded	1	1	dd
H _B	Shielded	2	1	td
H _C	Deshielded	2	1	td
H _D	Shielded	1	1	dd

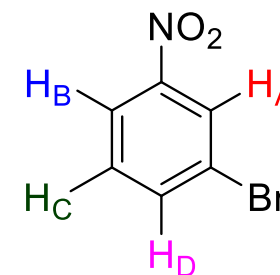


Aromatic Splitting patterns

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

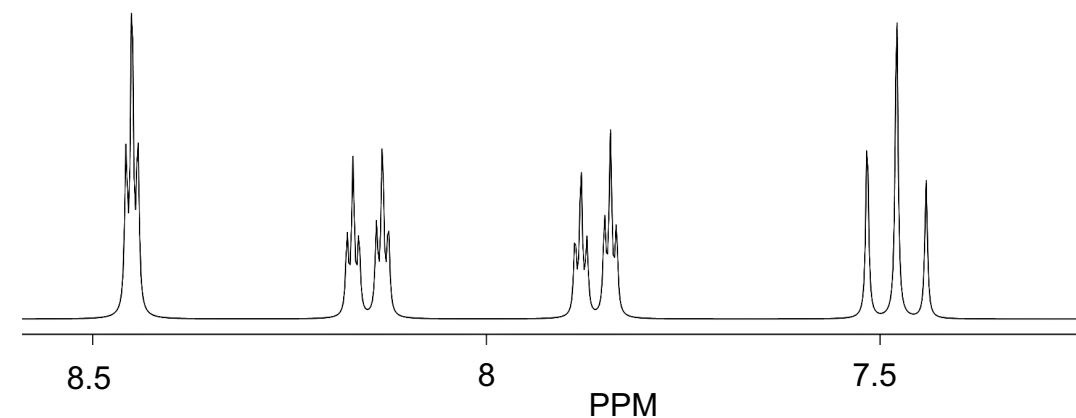
Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”



Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
------------------------	------------------------	------------------------	----------------------------

HA
HB
HC
HD

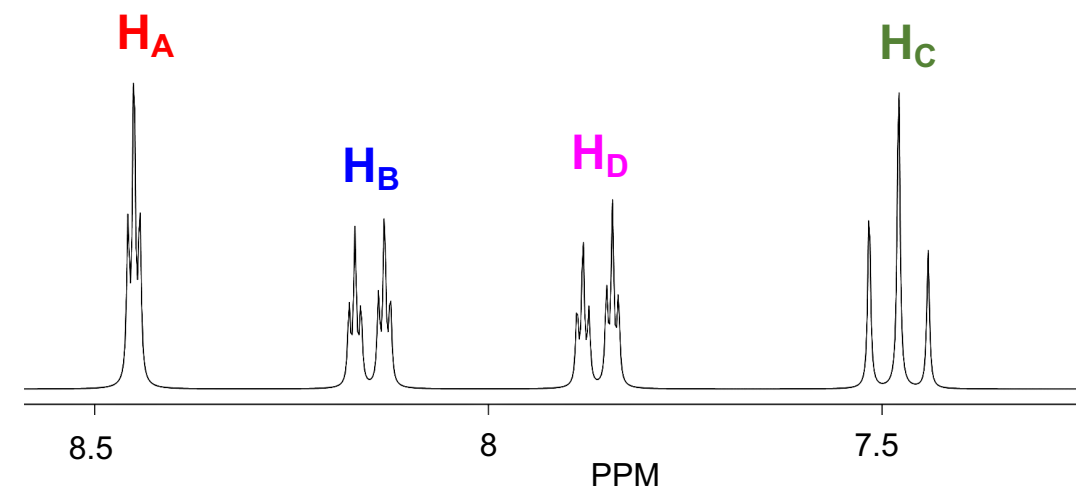
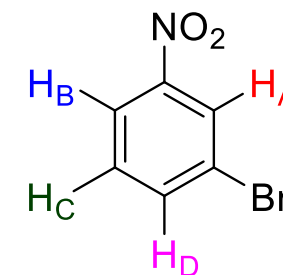


Aromatic Splitting patterns

$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$ is always true

Count all 3J coupled protons (**n**) and count all 4J coupled protons (**m**)

Aromatic splitting pattern will be “**n+1 of m+1**”



	Electronic environment	# of 3J protons (n)	# of 4J protons (m)	Expected splitting pattern
HA	*Deshielded	0	2	t (small J)
HB	*Deshielded	1	2	dt
HC	Neutral	2	0	t (large J)
HD	Deshielded	1	2	dt



Video Summary

4-bond coupling interactions are very common in aromatic rings

$$^3J_{\text{aromatic}} > ^4J_{\text{aromatic}}$$

Count all 3-bond coupling partners (**n**)

Count all 4-bond coupling partners (**m**)

- Aromatic splitting patterns are expected to be “**n+1 of m+1**”

*Assumes all $^3J_{\text{aromatic}}$ are equal and all $^4J_{\text{aromatic}}$ are equal

Next video: Diastereotopic Protons